

Лабораторная работа №1

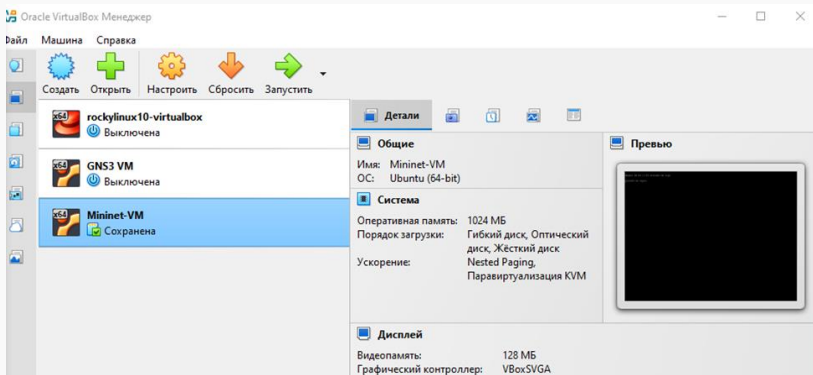
Введение в Mininet

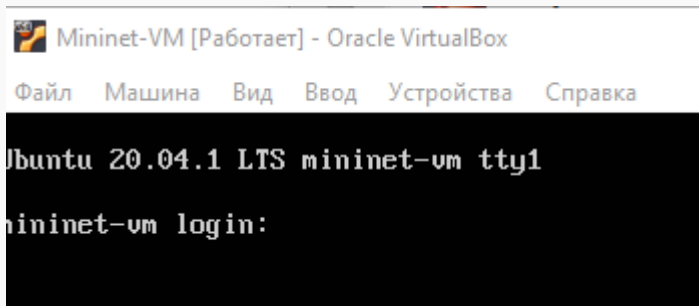
Владимир Базлов

11 сентября 2026

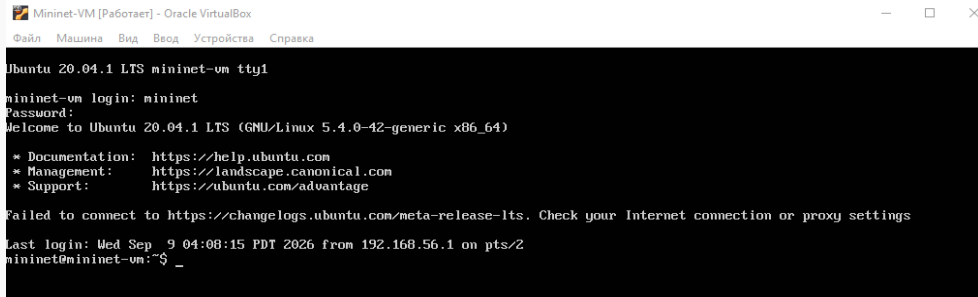
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Импорт виртуальной машины Mininet





Вход в систему



```
Mininet-VM [Работает] - Oracle VirtualBox
Файл  Машина  Вид  Ввод  Устройства  Справка

Ubuntu 20.04.1 LTS mininet-vm tty1

mininet-vm login: mininet
Password:
Welcome to Ubuntu 20.04.1 LTS (GNU/Linux 5.4.0-42-generic x86_64)

 * Documentation:  https://help.ubuntu.com
 * Management:    https://landscape.canonical.com
 * Support:       https://ubuntu.com/advantage

Failed to connect to https://changelogs.ubuntu.com/meta-release-lts. Check your Internet connection or proxy settings

Last login: Wed Sep  9 04:08:15 PDT 2026 from 192.168.56.1 on pts/2
mininet@mininet-vm:~$ _
```

Определение IP-адреса виртуальной машины

```
Last login: Wed Sep  5 01:00:13 PDT 2020 from 192.168.56.1 on pts/2
mininet@mininet-vm:~$ ifconfig
eth0: flags=4163<UP,BROADCAST,RUNNING,MULTICAST>  mtu 1500
    inet 192.168.56.3  netmask 255.255.255.0  broadcast 192.168.56.255
    ether 08:00:27:ef:40:79  txqueuelen 1000  (Ethernet)
    RX packets 3  bytes 1770 (1.7 KB)
    RX errors 0  dropped 0  overruns 0  frame 0
    TX packets 3  bytes 1026 (1.0 KB)
    TX errors 0  dropped 0 overruns 0  carrier 0  collisions 0

lo: flags=73<UP,LOOPBACK,RUNNING>  mtu 65536
    inet 127.0.0.1  netmask 255.0.0.0
    loop txqueuelen 1000  (Local Loopback)
    RX packets 160  bytes 12256 (12.2 KB)
    RX errors 0  dropped 0  overruns 0  frame 0
    TX packets 160  bytes 12256 (12.2 KB)
    TX errors 0  dropped 0 overruns 0  carrier 0  collisions 0

mininet@mininet-vm:~$
```

```
mininet@mininet-vm:~$ ssh -Y mininet@192.168.56.1
```

```
mininet@mininet-vm:~$ sudo mn
*** Creating network
*** Adding controller
*** Adding hosts:
h1 h2
*** Adding switches:
s1
*** Adding links:
(h1, s1) (h2, s1)
*** Configuring hosts
h1 h2
*** Starting controller
c0
*** Starting 1 switches
s1 ...
*** Starting CLI:
mininet>
```

```
mininet> nodes  
available nodes are:  
c0 h1 h2 s1  
mininet>
```

```
mininet> net
h1 h1-eth0:s1-eth1
h2 h2-eth0:s1-eth2
s1 lo: s1-eth1:h1-eth0 s1-eth2:h2-eth0
c0
mininet>
```

Проверка IP-адресов на хостах

```
mininet> h1 ifconfig
h1-eth0: flags=4163<UP,BROADCAST,RUNNING,MULTICAST> mtu 1500
    inet 10.0.0.1 netmask 255.0.0.0 broadcast 10.255.255.255
    ether ae:34:77:c4:75:8a txqueuelen 1000 (Ethernet)
    RX packets 0 bytes 0 (0.0 B)
    RX errors 0 dropped 0 overruns 0 frame 0
    TX packets 0 bytes 0 (0.0 B)
    TX errors 0 dropped 0 overruns 0 carrier 0 collisions 0

lo: flags=73<UP,LOOPBACK,RUNNING> mtu 65536
    inet 127.0.0.1 netmask 255.0.0.0
    loop txqueuelen 1000 (Local Loopback)
    RX packets 0 bytes 0 (0.0 B)
    RX errors 0 dropped 0 overruns 0 frame 0
    TX packets 0 bytes 0 (0.0 B)
    TX errors 0 dropped 0 overruns 0 carrier 0 collisions 0

mininet> h2 ifconfig
h2-eth0: flags=4163<UP,BROADCAST,RUNNING,MULTICAST> mtu 1500
    inet 10.0.0.2 netmask 255.0.0.0 broadcast 10.255.255.255
    ether b6:02:f6:e6:c5:c9 txqueuelen 1000 (Ethernet)
    RX packets 0 bytes 0 (0.0 B)
    RX errors 0 dropped 0 overruns 0 frame 0
    TX packets 0 bytes 0 (0.0 B)
    TX errors 0 dropped 0 overruns 0 carrier 0 collisions 0

lo: flags=73<UP,LOOPBACK,RUNNING> mtu 65536
    inet 127.0.0.1 netmask 255.0.0.0
    loop txqueuelen 1000 (Local Loopback)
    RX packets 0 bytes 0 (0.0 B)
    RX errors 0 dropped 0 overruns 0 frame 0
    TX packets 0 bytes 0 (0.0 B)
    TX errors 0 dropped 0 overruns 0 carrier 0 collisions 0

mininet>
```

```
mininet> h1 ping 10.0.0.2
PING 10.0.0.2 (10.0.0.2) 56(84) bytes of data.
 64 bytes from 10.0.0.2: icmp_seq=1 ttl=64 time=3.11 ms
 64 bytes from 10.0.0.2: icmp_seq=2 ttl=64 time=0.230 ms
 64 bytes from 10.0.0.2: icmp_seq=3 ttl=64 time=0.055 ms
 64 bytes from 10.0.0.2: icmp_seq=4 ttl=64 time=0.075 ms
 64 bytes from 10.0.0.2: icmp_seq=5 ttl=64 time=0.055 ms
 64 bytes from 10.0.0.2: icmp_seq=6 ttl=64 time=0.081 ms
 64 bytes from 10.0.0.2: icmp_seq=7 ttl=64 time=0.056 ms
 64 bytes from 10.0.0.2: icmp_seq=8 ttl=64 time=0.054 ms
 64 bytes from 10.0.0.2: icmp_seq=9 ttl=64 time=0.054 ms
 64 bytes from 10.0.0.2: icmp_seq=10 ttl=64 time=0.070 ms
 64 bytes from 10.0.0.2: icmp_seq=11 ttl=64 time=0.056 ms
 64 bytes from 10.0.0.2: icmp_seq=12 ttl=64 time=0.068 ms
 64 bytes from 10.0.0.2: icmp_seq=13 ttl=64 time=0.055 ms
 64 bytes from 10.0.0.2: icmp_seq=14 ttl=64 time=0.057 ms
 64 bytes from 10.0.0.2: icmp_seq=15 ttl=64 time=0.054 ms
 64 bytes from 10.0.0.2: icmp_seq=16 ttl=64 time=0.054 ms
 64 bytes from 10.0.0.2: icmp_seq=17 ttl=64 time=0.054 ms
 64 bytes from 10.0.0.2: icmp_seq=18 ttl=64 time=0.066 ms
 64 bytes from 10.0.0.2: icmp_seq=19 ttl=64 time=0.054 ms
 64 bytes from 10.0.0.2: icmp_seq=20 ttl=64 time=0.071 ms
 64 bytes from 10.0.0.2: icmp_seq=21 ttl=64 time=0.088 ms
 64 bytes from 10.0.0.2: icmp_seq=22 ttl=64 time=0.066 ms
 64 bytes from 10.0.0.2: icmp_seq=23 ttl=64 time=0.061 ms
```

Итоги работы

В ходе лабораторной работы были выполнены следующие действия:

- развёрнута виртуальная машина Mininet в системе виртуализации VirtualBox;
- выполнено подключение к виртуальной машине по SSH с переадресацией X11;
- запущена минимальная топология Mininet командой `sudo mn`;
- изучены основные команды интерфейса командной строки Mininet: `nodes`, `net`, `ifconfig`, `ping`, `exit`;
- проверена связность между хостами h1 и h2 (IP-адреса 10.0.0.1 и 10.0.0.2).